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Complexity Adjusted Comparisons of LPs and VARs: Evidence from Monte-Carlo Simulations

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1. Introduction (1.5-2 pages)

To infer statements about the dynamic response to exogenous shocks, the estimation of the corresponding impulse response has relied on two fundamental approaches, either through vector autoregressive (VAR) or Local Projection (LP) models. Since the introduction of the latter, an extensive literature has emerged discussing the relative merits of the two. A first strand of literature has established that, under relatively general conditions, LPs and VARs estimate the same population impulse responses (Plagborg-Møller & Wolf, 2021; Fusari et al. 2026). However, in the discussion of finite sample behavior, LPs were often viewed as less sensitive to dynamic misspecification whereas VARs were seen as potentially more statistically efficient (Li et al., 2024; Montiel Olea et al., 2025). Ludwig (2026) offers an analytical approach of explaining the observed finite sample bias-variance trade-off. He poses that LPs can be represented as a sequence of VAR estimations increasing in lag length as the estimated horizon increases and thus argues for a renewed thinking about how we can compare LPs and VARs – not by lag length, but by effective complexity.

The first part of this paper aims to build on the discussion outlined above by asking whether conventional finite-sample conclusions are robust to alternative benchmark comparisons that relax the standard equal-lag restriction. To answer this question, the paper operationalizes Ludwig’s (2026) central implication through a complexity-adjusted benchmarking exercise in which a fixed LP estimator is compared to progressively higher-order VAR specifications that proxy increasing effective dynamic flexibility (De Graeve & Westermarck, 2025). This comparison is conducted across empirically relevant sample sizes (see e.g. Herbst & Johannsen 2021) to assess how the resulting bias-variance trade-off depends on the available data.

The central research question is: Do LPs retain their apparent robustness advantage once they are compared not only to equal-lag VARs, but also to higher-order VARs that approximate the effective complexity implied by LP estimation? The contribution is not to establish a new LP-VAR equivalence result, but to ask whether standard Monte Carlo comparisons between LPs and equal-lag VARs are affected by the fact that LPs correspond to a more flexible dynamic benchmark.

The second part of the paper introduces controlled dynamic misspecification. While the baseline experiment studies complexity-adjusted LP-VAR comparisons under correctly specified linear dynamics, the main practical argument for LPs concerns their robustness when the estimated linear model is only an approximation to the true DGP (Montiel Olea et al., 2021). To study this channel explicitly, the paper introduces a controlled degree of misspecification into the DGP and examines how the relative performance of LPs, equal-lag VARs, and higher-order VARs changes as misspecification becomes more severe.

The simulation design proceeds in three steps. First, we consider a correctly specified bivariate VAR(4) DGP, allowing LP(4), VAR(4), and higher-order VARs to be compared under common autoregressive information. Second, we introduce dynamic misspecification by adding a moving-average shock component, thereby generating VARMA(4,1) dynamics while continuing to estimate finite-order VARs and LPs. Third, we introduce nonlinear misspecification through quadratic terms in the DGP while retaining linear estimators. This structure separates the role of dynamic approximation error from nonlinear functional-form misspecification and allows us to study whether conventional LP-VAR finite-sample rankings are robust to complexity-adjusted VAR benchmarks.

For each estimator, horizon, DGP, and sample size, we compute bias, variance, mean squared

error, and empirical coverage rates. Bias and variance describe the finite-sample trade-off in point estimation, while coverage assesses whether the corresponding inference procedures re-main reliable under increasing dynamic or nonlinear misspecification.

2. Theoretical Background (total: 5-6.5 pages)

Impulse response functions can be nonparametrically defined as the difference between two conditional forecasts of the outcome variable (e.g. Koop (1996)). Both forecasts are conditioned on the same history up to period $t-1$, but differ in the realization at time t . While one conditions on a shock of size δ to the variable of interest, the other one does not. Following Ramey (2016) shocks hereby refers to a significant exogenous force representing unanticipated movements in exogeneous variables, that is uncorrelated with other current and lagged control variables and other exogeneous shocks. The observed difference in the projected paths at horizon h then identifies the causal effect of the shock in population:

$$\text{IR}(h, \delta) \equiv \mathbb{E}[y_{t+h}|s_t = s_0 + \delta; \mathbf{x}_t] - \mathbb{E}[y_{t+h}|s_t = s_0; \mathbf{x}_t] \quad (1)$$

Vector Autoregressions and Local Projections serve as two complementary approaches to estimating impulse responses in finite samples. The former, introduced by Sims (1980), parametrizes the joint dynamics of all variables in a single autoregressive system and impulse responses are then obtained as iterated forecasts from the estimated companion matrix. The latter, established by Jordà (2005), takes a direct forecasting approach by estimating the responses at each horizon independently via OLS, projecting the future outcome directly onto current and lagged values of the data. The two approaches compound estimation error differently across horizons and, at equal lag order, occupy opposite ends of a bias-variance spectrum in finite samples (Li et al., 2024; Olea et al., 2025). Local projections tend toward higher variance, vector autoregressions toward higher bias under misspecification. More recently, however, Ludwig (2026) shows that the equal-order comparison is itself misleading. At a given lag order p , $\text{LP}(p)$ is strictly more general than $\text{VAR}(p)$, so the conventional bias-variance trade-off conflates the choice of estimator with the choice of model complexity.

2.1 Introduction to VARS (1.5 - 2 pages)

Following Lüthkepohl (2005), we start by defining a zero mean $\text{VAR}(p)$ model of the form where the outcome of interest, y_t , is a $K \times 1$ random vector, A_i are fixed $K \times K$ coefficient matrices and u_t are a K dimensional white noise processes with $\mathbb{E}[u_t] = 0$, $\mathbb{E}[u_t u_t'] = \Sigma_u$, i.e. being a nonsingular covariance matrix, and $\mathbb{E}[u_t u_s'] = 0$ for $t \neq s$

$$y_t = \nu + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t, \quad t = 0, \pm 1, \pm 2, \dots \quad (2)$$

or simply in form of the companion $\text{VAR}(1)$

$$Y_t = \nu + \mathbf{A} Y_{t-1} + U_t \quad (3)$$

where

$$Y_t := \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}, \nu := \begin{bmatrix} \nu \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{A} := \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}, U_t := \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Assuming Y_t satisfies the stability condition such that all roots, z , of the characteristic equation lie strictly outside the complex unit circle, the process is covariance stationary and admits a Wold moving-average representation.

$$\det(I_{Kp} - \mathbf{A}z) \neq 0 \text{ for } |z| \leq 1 \iff Y_t = \sum_{i=0}^{\infty} \mathbf{A}^i U_{t-i} \quad (4)$$

The Wold moving-average coefficients for a specific horizon h , $\Psi_h \equiv [\mathbf{A}^h]$ are obtained by noting that since $U_{t-i} = (u_{t-i}, 0, \dots, 0)'$ has nonzero entries only in its first K positions. The effect of u_{t-i} on y_t is therefore fully captured by the upper-left $K \times K$ block of $\mathbf{A}^i$. Extracting its upper-left $K \times K$ block thus yields Ψ_h corresponding to the impulse response of y_t to a unit reduced-form innovation h periods ago. However, since the components of u_t are generally contemporaneously correlated, $\Sigma_u = \mathbb{E}(u_t u_t') \neq I_K$, a unit shock to one component of u_t will in general be accompanied by simultaneous movements in the other components. Consequently, the reduced-form impulse responses Ψ_h do not admit a structural interpretation. Structural identification is only achieved by imposing restrictions on the impact matrix B in the decomposition $u_t = B\varepsilon_t$, where $\varepsilon_t \sim (0, I_K)$ are mutually uncorrelated structural shocks.

The most common approach is recursive Cholesky identification, which restricts B to be lower triangular with positive diagonal entries, uniquely recovering B as the Cholesky factor of $\Sigma_u = BB'$. The recursive ordering imposes a causal structure on the system: the variable ordered first responds contemporaneously to all shocks, while the variable ordered last is insulated from all contemporaneous feedback. Defining the orthogonalised innovations $w_t := B^{-1}u_t$ yields $\mathbb{E}(w_t w_t') = B^{-1}\Sigma_u(B^{-1})' = I_K$, so that the components of w_t are mutually uncorrelated with unit variance. Substituting into the Wold representation gives the structural moving-average representation

$$y_t = \sum_{j=0}^{\infty} \Theta_j w_{t-j}, \quad \Theta_j := \Psi_j B, \quad (5)$$

where the jk -th element of Θ_h measures the response of variable j to a unit innovation in the k -th orthogonalised shock h periods ago. Selecting the response of variable i to the j -th shock yields the structural impulse response at horizon h ,

$$\theta_h^{ij} = e_i' \Psi_h B e_j, \quad (6)$$

where $e_i, e_j \in \mathbb{R}^K$ are standard basis vectors.

2.1.1 POINT ESTIMATION

Estimation of the structural impulse response θ_h^{ij} proceeds in two stages. In the first stage, the reduced-form VAR is estimated equation by equation by ordinary least squares. Since every equation contains the same set of regressors—a constant and p lags of all variables—generalised least squares offers no efficiency gain over equation-by-equation OLS, and the two estimators coincide **zellner1962efficient**. Collecting the coefficients in $\hat{A} = (\hat{A}_1, \dots, \hat{A}_p)$, the OLS estimator is consistent and asymptotically normal under the stability condition,

$$\sqrt{T} (\text{vec}(\hat{A}) - \text{vec}(A)) \xrightarrow{d} \mathcal{N}(0, \Gamma^{-1} \otimes \Sigma_u), \quad (7)$$

where $\Gamma = \mathbb{E}(Z_t Z_t')$ is the second-moment matrix of the stacked regressors $Z_t = (1, y'_{t-1}, \dots, y'_{t-p})'$ **lutkepohl2005**. The innovation covariance matrix is estimated by the residual analogue $\hat{\Sigma}_u = (T - Kp - 1)^{-1} \sum_t \hat{u}_t \hat{u}_t'$. The lag order p is selected by an information criterion such as the AIC **akaike1974new** or the BIC **schwarz1978estimating**, which balance in-sample fit against model complexity.

In the second stage, the structural objects are recovered from the reduced-form estimates. The impact matrix $\hat{B}$ is obtained as the Cholesky factor of $\hat{\Sigma}_u$, and the reduced-form impulse responses $\hat{\Psi}_h$ are computed from $\hat{A}$ via the recursion

$$\hat{\Psi}_h = \sum_{\ell=1}^{\min(h,p)} \hat{A}_\ell \hat{\Psi}_{h-\ell}, \quad \hat{\Psi}_0 = I_K, \quad (8)$$

or equivalently as the upper-left block of $\hat{\mathbf{A}}^h$. The structural impulse response estimator is then

$$\hat{\theta}_h^{ij} = e_i' \hat{\Psi}_h \hat{B} e_j, \quad (9)$$

a smooth nonlinear function of the first-stage estimates $\hat{A}$ and $\hat{\Sigma}_u$.

ASYMPTOTIC INFERENCE

Because the structural impulse response $\hat{\theta}_h^{ij}$ is a smooth nonlinear function of the first-stage estimates, its asymptotic distribution follows from the delta method. Let $\alpha = \text{vec}(A_1, \dots, A_p)$ and $\sigma = \text{vech}(\Sigma_u)$ collect the reduced-form parameters, with the joint estimator satisfying

$$\sqrt{T} \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\sigma} - \sigma \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{pmatrix} \Sigma_\alpha & 0 \\ 0 & \Sigma_\sigma \end{pmatrix} \right). \quad (10)$$

Since the impulse response is obtained through the differentiable mapping $\theta_h^{ij} = g(\alpha, \sigma)$, the delta method propagates this asymptotic normality through the Jacobian of g , yielding

$$\sqrt{T} (\hat{\theta}_h^{ij} - \theta_h^{ij}) \xrightarrow{d} \mathcal{N} \left(0, \frac{\partial g}{\partial \alpha'} \Sigma_\alpha \frac{\partial g'}{\partial \alpha} + \frac{\partial g}{\partial \sigma'} \Sigma_\sigma \frac{\partial g'}{\partial \sigma} \right), \quad (11)$$

where the Jacobian of the reduced-form responses $G_h = \partial \text{vec}(\Psi_h) / \partial \alpha'$ has the closed form $G_h = \sum_{m=0}^{h-1} J(\mathbf{A}')^{h-1-m} \otimes \Psi_m$ **lutkepohl2005**. The resulting standard errors deliver confidence intervals

for $\hat{\theta}_h^{ij}$ that are valid pointwise in the horizon h , provided the process is stationary with roots bounded away from the unit circle.

This pointwise validity comes with an important caveat. As **lutkepohl2005** notes, the asymptotic covariance matrix in the proposition can become singular for certain parameter configurations, in which case the usual t -ratios are no longer asymptotically standard normal and inference based on them may be misleading. More fundamentally, the delta-method approximation is valid only pointwise: it is not guaranteed to hold uniformly over the persistence of the process, nor at horizons that grow with the sample size **mikusheva2012onedimensional**. These qualifications motivate a closer look at the finite-sample behaviour of VAR-based inference.

Small-Sample Limitations

The favourable asymptotic properties established above are qualified by three interrelated problems that arise in the sample sizes typical of macroeconomic applications. All three originate in the same feature that gives the VAR its parsimony: impulse responses are obtained by iterating the estimated one-step dynamics forward, so that estimation error enters the response nonlinearly and accumulates across horizons.

First, estimation error propagates through iteration. Because $\hat{\Psi}_h$ is the upper-left block of $\hat{\mathbf{A}}^h$, error in $\hat{A}$ enters the impulse response through powers of the companion matrix and is compounded multiplicatively as the horizon grows. Even small coefficient biases therefore translate into substantial distortions of the estimated responses at longer horizons **kilian1998small**. In the limiting case of unit or near-unit roots, **phillips1998impulse** shows that long-horizon impulse responses from unrestricted VARs are not even consistently estimated, converging instead to random variables rather than the true responses.

Second, the autoregressive coefficients themselves are biased downward in short samples. It is a long-established result that least squares underestimates the persistence of an autoregressive process **pope1990biases**, and because $\hat{\Psi}_h$ inherits this bias through the recursion, the estimated impulse responses decay too quickly and understate the true persistence of the system. The distortion is most severe precisely when persistence is high and the sample is short—the empirically relevant case for many macroeconomic series.

Third, these distortions undermine delta-method inference. When the largest roots approach the unit circle, the asymptotic normal approximation deteriorates and the resulting confidence intervals undercover, increasingly so at longer horizons **kilian1998small**. The failure is not merely a finite-sample inconvenience but reflects the non-uniformity of the underlying asymptotics: the normal approximation holds pointwise but not uniformly over the persistence parameter **mikusheva2012onedimensional**. The standard practical remedy is the bias-corrected bootstrap of **kilian1998small**, which bias-adjusts the estimated coefficients before resampling and thereby accounts for both the bias and the skewness of the small-sample impulse response distribution.

Taken together, these limitations—all rooted in the iterated-forecasting structure of the VAR—motivate the study of estimators that recover impulse responses directly at each horizon, without iterating a fitted one-step model forward. The local projection approach of the following section is precisely such an alternative.

Asymptotic Inference: Lütkepohl (1990), Phillips (1998)

Small-sample limitations: Bias propagation through iterated forecasting:

Downward bias in autoregressive samples:

Failure of delta-method inference under high persistence and long horizon Kilian (1998)

Kilian (1999) Kilian & Chang (2000)

2.2 Introduction to Local Projections (1.5 -2 pages)

Whereas the vector autoregression recovers the impulse response function by first estimating the full system dynamics and then iterating the fitted model forward, the local projection approach estimates the same object directly, horizon by horizon, through a sequence of OLS regressions.

Assuming orthogonality of the shock with the projection error term and a linear projection model, we can obtain the impulse response to the shock at time h using the following OLS model:

$$y_{t+h} = \quad (12)$$

2.3 On the Equivalence of VARs and Local Projections (2-2.5 pages)

As Local Projection inference gained in popularity over the last couple of years, more and more scholars have started comparing the asymptotical and finite sample trade-offs of both methods. Prior research has established the notion that SVARs are computationally more efficient while LPs inherit favorable robustness to misspecification (see e.g. Ramey, 2016; Jordà, 2005). The notion applies in analogy to the comparison of iterated vs. direct forecasting as established by Marcellino et al. (2006).

Plagborg-Møller & Wolf (2021) were the first to actually show that under weak stationarity and unrestricted lag structure both methods estimate the identical impulse response function in population. Therefore any conventional LP impulse response can be obtained through an appropriately ordered recursive VAR and any VAR impulse response through a LP with appropriate control variables. - they do not impose restrictions of the underlying DGP The logic applies that any VAR model with sufficiently large lag lengths is able to capture the whole covariance properties of the data such that $\text{VAR}(\infty)$ forecasts coincide with direct LP forecasts. The authors come to the conclusion: - finite lag length, impulse response approximately agree if the same lag length is used for both but differ for $h \geq p$. - in population, linear local projections are exactly as robust to non-linearities in the DGP as linear VARs

Here I will discuss the theoretical results that establish the equivalence of VARs and LPs under certain conditions, as shown by Plagborg-Møller & Wolf (2021) and Fusari et al. (2026). I will also explain how this equivalence is affected by finite sample properties and dynamic misspecification, as discussed by Li et al. (2024), Montiel Olea et al. (2025), and Ludwig (2026).

The conventional finite-sample comparison between local projections and vector autoregressions is typically conducted at equal lag order, pitting $\text{LP}(p)$ against $\text{VAR}(p)$. Ludwig (2026) shows that this convention rests on a misleading notion of parsimony. His Theorem 1 establishes an exact finite-sample identity: the h -step prediction of an $\text{LP}(p)$ is the forward iteration of a sequence of one-step VAR predictions whose lag order rises from p to $p + h - 1$. An equal-order $\text{VAR}(p)$ is therefore a restricted special case of $\text{LP}(p)$, obtained by holding the lag order fixed along the recursion rather than allowing it to increase. The familiar empirical pattern—lower approximation bias but higher sampling variance for LP relative to a same-order VAR—is, from this vantage point, largely a mechanical consequence

of comparing a more general specification with its own restriction, rather than evidence of an intrinsic property of either estimator. This matters because it leaves the standard reading of the bias-variance trade-off, as articulated by Plagborg-Møller and Wolf (2021) and Montiel Olea et al. (2025), confounded: at equal lag order, the method effect (LP versus VAR) cannot be separated from the complexity effect (more versus fewer effective autoregressive parameters).

3. Baseline Simulation (4-5 pages)

This confounding motivates the first part of our study. Because Ludwig’s contribution is analytical and deliberately abstains from a simulation-based horse race, the question of whether the conventional finite-sample conclusions survive once the equal-lag restriction is relaxed remains open. We therefore operationalize Ludwig’s central implication through a complexity-adjusted benchmarking exercise that holds parsimony more nearly fixed, comparing a fixed LP(4) estimator against progressively higher-order VAR specifications that proxy its increasing effective dynamic flexibility. Crucially, because the identity in Theorem 1 maps LP(p) to a sequence of VARs rather than to any single higher-order VAR, no individual VAR is the appropriate counterpart; instead, LP(4) is read as a reference line against the full VAR order frontier, against which its bias, variance, and coverage can be located. We frame the exercise as a test of robustness rather than as a foregone conclusion: the aim is to assess whether the conventional bias-variance verdict persists, dissolves, or reverses once the comparison is placed on more comparable terms, and to identify the features of the data-generating process under which each procedure is favored.

3.1 Simulation Design

AIM: This simulation study examines whether conventional finite-sample conclusions about the bias-variance tradeoff between Local Projections and Vector Autoregressions are robust to relaxing the equal-lag restriction. Specifically, we compare a fixed LP(4) estimator against a sweep of VAR specifications of increasing order, ranging from VAR(1) to VAR(p+H-1), thereby constructing a complexity-adjusted benchmark in the spirit of Ludwig (2026). The central question is whether LP(4) retains its finite-sample MSE advantage over the equal-lag VAR(4) once the VAR is allowed to exploit its natural complexity advantage at longer horizons.

DGP: The baseline DGP is a stable bivariate zero-mean VAR(4) with recursive Cholesky identification. Recursive identification refers to the lower-triangular structure of the impact matrix B , obtained via Cholesky decomposition of the reduced-form error covariance matrix $\Sigma_u = BB'$. This imposes a contemporaneous zero restriction: the first variable does not respond on impact to shocks in the second variable, while the second variable may respond to both structural shocks simultaneously.

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + u_t, \quad u_t \sim (0, \Sigma_u), \quad y_t \in \mathbb{R}^2 \quad (13)$$

$$\Sigma_u = BB', \quad B = \begin{pmatrix} b_{11} & 0 \\ b_{21} & b_{22} \end{pmatrix}, \quad b_{11}, b_{22} > 0 \quad (14)$$

$$\rho(\mathcal{A}) < 1, \quad \mathcal{A} = \begin{pmatrix} A_1 & A_2 & A_3 & A_4 \\ I_2 & 0 & 0 & 0 \\ 0 & I_2 & 0 & 0 \\ 0 & 0 & I_2 & 0 \end{pmatrix} \in \mathbb{R}^{8 \times 8} \quad (15)$$

Persistence is calibrated via the spectral radius of the companion matrix across the empirically relevant range $\rho \in (0.5, 0.95)$. The lower bound reflects the observation that, for weakly persistent processes, impulse responses decay rapidly toward zero at longer horizons, rendering the bias-variance comparison between LP and VAR uninformative. The upper bound approaches the unit-root boundary while maintaining covariance stationarity, covering the high-persistence region where finite-sample differences between estimators are most pronounced.

$$A_1 \leftarrow \frac{\rho}{\rho(\mathcal{A})} \cdot A_1, \quad \rho \in (0.5, 0.95) \quad (16)$$

Sample sizes are set to $T \in \{100, 200, 240, 500\}$, covering the range of quarterly macroeconomic samples typically encountered in applied work. The choice follows Herbst & Johannsen (2021), who document that the modal sample size in the structural VAR literature lies between 100 and 240 quarterly observations. The value $T = 500$ serves as a large-sample reference point to assess convergence toward the population equivalence result of Plagborg-Møller & Wolf (2021).

The structural estimand is $\theta_h = \Psi_h B e_1$, where Ψ_h denotes the h -step ahead moving-average matrix of the reduced-form VAR, defined via $y_{t+h} = \sum_{j=0}^{\infty} \Psi_j u_{t+h-j}$ with $\Psi_0 = I_2$, B is the Cholesky impact matrix, and $e_1 = (1, 0)'$ is a selection vector. The primitive shock is normalized as a unit shock: we trace the dynamic response of the system to a one-standard-deviation increase in the first structural innovation ε_{1t} . Since B is normalized such that $\text{Var}(\varepsilon_t) = I_2$, no further rescaling is required.

$$u_t = B \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2) \quad (17)$$

The innovation term is specified as Gaussian white noise. The reasoning behind Gaussianity is two-fold. First, our primary performance measures are bias, variance, and RMSE of the IRF estimates, which in population depend exclusively on the second moments of the underlying process, i.e. the VAR coefficients and the covariance matrix Σ_u . The distribution of the innovation term is therefore irrelevant for the point estimator comparison. Second, we follow Plagborg-Møller & Wolf (2021), who introduce Gaussianity explicitly as an expositional device that allows conditional expectations to be written in place of linear projections. As all results hold under covariance stationarity alone, the distributional assumption leaves the analysis qualitatively unchanged. Importantly, this choice is orthogonal to the misspecification parameters introduced in the extensions: varying the MA parameter θ in Extension 1 or the nonlinearity parameter γ in Extension 2 alters the structure of the DGP while leaving the innovation distribution fixed, ensuring that differences in estimator performance across designs are attributable solely to the respective misspecification channel.

ESTIMAND:

$$\theta_h = \Psi_h B e_1, \quad e_1 = (1, 0)', \quad h = 1, \dots, 20 \quad (18)$$

METHODS: We compare two classes of estimators. The first is a fixed LP(4) estimator, which serves as the reference point. At each horizon $h = 1, \dots, H$, the LP(4) estimator is obtained from the

OLS regression

$$y_{1,t+h} = \mu_h + \beta_h x_t + \sum_{\ell=1}^4 \delta'_{h,\ell} y_{t-\ell} + \xi_{h,t}, \quad h = 1, \dots, H \quad (19)$$

where $x_t = e'_1(\hat{B}^{(4)})^{-1}\hat{u}_t^{(4)}$ is the estimated structural shock recovered from the first-stage VAR(4). The structural impulse response estimate at horizon h is $\hat{\theta}_h^{\text{LP}} = \hat{\beta}_h$.

The second class consists of a sweep of VAR estimators of increasing order, VAR(q) for $q = 1, \dots, p + H - 1$, where $p = 4$ and $H = 20$. For each order q , the VAR is estimated equation-by-equation via OLS, the reduced-form covariance matrix $\hat{\Sigma}_u^{(q)}$ is Cholesky-decomposed to recover the structural impact matrix $\hat{B}^{(q)}$, and the h -step ahead structural IRF is computed as

$$\hat{\theta}_h^{\text{VAR}(q)} = \hat{\Psi}_h^{(q)} \hat{B}^{(q)} e_1 \quad (20)$$

where $\hat{\Psi}_h^{(q)}$ denotes the h -step ahead moving-average matrix of the estimated VAR(q). This sweep constitutes the complexity-adjusted benchmark motivated by Ludwig (2026): by Theorem 1 of that paper, LP(4) at horizon h is algebraically equivalent to a sequence of VAR predictions stepping through orders $4, 5, \dots, 4 + h - 1$. The fair comparison is therefore not LP(4) against VAR(4) alone, but against the full range of VAR specifications up to order $p + H - 1$. Note that this equivalence holds exactly for the sequence of VARs of increasing order; comparing LP(4) to any single VAR(q^*) is an approximation, with the complexity-adjusted single-order approximation given by $q^* = p + \frac{h-1}{2}$ (Ludwig, 2026). The equal-lag VAR(4) is nested in the sweep as a special case and is retained as an additional reference point to connect to the conventional benchmark in the literature.

Identification of the structural shock is implemented as follows. For each VAR(q) in the sweep, the structural impact matrix $\hat{B}^{(q)}$ is obtained via Cholesky decomposition of the order-specific residual covariance matrix $\hat{\Sigma}_u^{(q)}$. For the LP(4), which does not estimate a complete system, identification is achieved through a first-stage VAR(4): the structural shock is constructed as $\hat{\varepsilon}_{1t} = e'_1(\hat{B}^{(4)})^{-1}\hat{u}_t^{(4)}$, consistent with the LP lag order $p = 4$. Since Plagborg-Møller & Wolf (2021) establish that LP and VAR share the same implicit shock in population under recursive identification, this ensures that any divergence in estimated IRFs reflects differences in propagation rather than identification. No data-dependent lag selection is performed within the sweep; all VAR orders $q = 1, \dots, p + H - 1$ are estimated on each simulated sample, ensuring that the comparison is across fixed specifications rather than across selection procedures.

PERFORMANCE: Performance is assessed along two dimensions, reported horizon-by-horizon for $h = 1, \dots, 20$. For point estimation, we report bias, variance, MSE, and RMSE of the IRF estimates $\hat{\theta}_h$ relative to the true estimand θ_h , computed across $B = 5,000$ Monte Carlo replications. The number of replications exceeds the $B = 1,000$ used by Li et al. (2024) and Olea et al. (2025), ensuring tighter Monte Carlo standard errors at the cost of additional computation. Formally:

$$\widehat{\text{Bias}}_h = \frac{1}{B} \sum_{b=1}^B \left(\hat{\theta}_h^{(b)} - \theta_h \right) \quad (21)$$

$$\widehat{\text{Var}}_h = \frac{1}{B-1} \sum_{b=1}^B \left(\hat{\theta}_h^{(b)} - \bar{\hat{\theta}}_h \right)^2 \quad (22)$$

$$\widehat{\text{MSE}}_h = \frac{1}{B} \sum_{b=1}^B \left(\hat{\theta}_h^{(b)} - \theta_h \right)^2 \quad (23)$$

$$\widehat{\text{RMSE}}_h = \sqrt{\widehat{\text{MSE}}_h} \quad (24)$$

Monte Carlo standard errors are reported alongside all point-estimator performance measures to quantify simulation uncertainty, with $\text{MCSE}(\widehat{\text{Bias}}_h) = \sqrt{\widehat{\text{Var}}_h/B}$.

For inference, we assess coverage of nominal 95% confidence intervals. For the VAR(q) sweep, confidence intervals are constructed via the delta method, propagating estimation uncertainty in the reduced-form coefficients $(\hat{A}_1, \dots, \hat{A}_q)$ through the nonlinear mapping to structural IRFs $\hat{\theta}_h^{\text{VAR}(q)} = \hat{\Psi}_h^{(q)} \hat{B}^{(q)} e_1$ (Kilian & Lütkepohl, 2017). For LP(4), confidence intervals are based on heteroskedasticity-robust standard errors, which suffice under the martingale difference assumption on the structural shock (Plagborg-Møller & Wolf, 2021). Coverage is defined as the fraction of replications in which the nominal 95% interval contains the true estimand θ_h :

$$\widehat{\text{Coverage}}_h = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \left[\theta_h \in \hat{CI}_h^{(b)} \right] \quad (25)$$

3.2 Simulation Results

4. Extension 1: Dynamic Misspecification (3-4 pages)

4.1 Simulation Design

Extension 1 examines whether the finite-sample MSE ranking between LP(4) and the complexity-adjusted VAR sweep is robust to dynamic misspecification of the Wold representation. The baseline VAR(4) is augmented by a first-order moving average component, yielding a VARMA(4,1) DGP:

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + u_t + \Theta u_{t-1}, \quad u_t = B \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2) \quad (26)$$

where $\Theta = \theta I_2$ is a scalar-scaled MA impact matrix, with misspecification strength $\theta \in \{0.1, 0.3, 0.6\}$. The case $\theta = 0$ nests back to the baseline VAR(4), ensuring continuity across designs. The VAR coefficient matrices $A_1, \dots, A_4$ and the impact matrix B are inherited from the baseline DGP, so that all variation across Extension 1 scenarios is attributable exclusively to the MA component.

The VARMA(4,1) DGP is invertible for all $\theta \in \{0.1, 0.3, 0.6\}$, admitting an infinite-order VAR representation. This means the complexity-adjusted VAR sweep retains its theoretical motivation: higher-order VAR(q) specifications progressively approximate the true dynamics as $q \rightarrow \infty$, though

at the cost of increasing parameter proliferation in finite samples. The central question is therefore whether LP(4) — which is robust to this misspecification by virtue of its zero asymptotic bias (Olea et al., 2025) — retains its MSE advantage over the equal-lag VAR(4), and where on the complexity-adjusted VAR frontier it sits relative to the baseline.

Persistence is varied across $\rho \in (0.5, 0.95)$ as in the baseline, since the degree to which the MA component distorts finite-sample VAR approximations depends directly on the persistence of the underlying process: at high ρ , shocks decay slowly and the MA term remains influential at longer horizons, amplifying the VAR’s approximation bias. Sample sizes follow the baseline design, $T \in \{100, 200, 240, 500\}$. The misspecification strength $\theta \in \{0.1, 0.3, 0.6\}$ is thus varied jointly with persistence and sample size, yielding a three-way design that isolates the marginal contribution of MA misspecification from the persistence and finite-sample channels already documented in the baseline.

The structural estimand $\theta_h = \Psi_h^{\text{VARMA}} B e_1$ is defined with respect to the true VARMA moving-average matrices Ψ_h^{VARMA} , computed analytically from the VARMA companion representation prior to each simulation run. Both estimators — LP(4) and the VAR(q) sweep — are applied without modification to the VARMA-generated data, so that the misspecification is fully absorbed into the finite-sample performance of each estimator rather than accommodated by the estimation procedure. Performance measures are identical to the baseline.

4.2 Simulation Results

5. Extension 2: Functional Misspecification (3-4 pages)

5.1 Simulation Design

Extension 2 examines whether the finite-sample MSE ranking between LP(4) and the complexity-adjusted VAR sweep is robust to functional misspecification of the DGP. The baseline VAR(4) is augmented by a quadratic term in the first structural shock, yielding a nonlinear DGP:

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + u_t + \gamma \cdot c(\varepsilon_{1,t}), \quad u_t = B \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2) \quad (27)$$

where $c(\varepsilon_{1,t}) = (0, \varepsilon_{1,t}^2 - 1)'$ is a mean-zero quadratic term in the first structural shock entering the second equation only, and $\gamma \in \{0.3, 0.6\}$ controls the degree of nonlinearity. The centering $\varepsilon_{1,t}^2 - 1$ ensures that the nonlinear term has mean zero, so that $\gamma = 0$ nests back to the baseline VAR(4).

This shock-quadratic specification is adopted in place of a level-quadratic or lag-quadratic formulation for two reasons. First, a quadratic term in lagged levels introduces positive feedback into the system dynamics and is explosive for moderate values of γ , violating the covariance stationarity required for the estimand θ_h to be well-defined. Second, and more importantly, a quadratic term in the structural shock isolates the nonlinearity in the transmission mechanism rather than in the propagation dynamics. This distinction matters for the research question: since the Cholesky-identified structural shock $\varepsilon_{1,t}$ enters the system linearly in the baseline VAR(4), a shock-quadratic extension targets precisely the channel emphasized by Plagborg-Møller & Wolf (2021) — namely, that linear LP and linear VAR are both blind to nonlinearities in the shock transmission in population (Proposition 1, p. 960). A lag-quadratic specification would instead distort the propagation dynamics and complicate

the identification of the structural shock, confounding the comparison with the baseline DGP. The shock-quadratic formulation thus provides the cleanest test of whether the complexity-adjusted VAR sweep offers any finite-sample advantage when functional misspecification is present but structurally uninvertible — that is, not approximable by higher-order linear VARs regardless of lag length.

The key theoretical motivation for this extension follows from Plagborg-Møller & Wolf (2021), who establish that linear LP and linear VAR estimate the same best linear approximation to the true IRF in population, regardless of whether the DGP is linear (Proposition 1, p. 960). Under equal lag length, both estimators are therefore equally blind to the quadratic misspecification in population, and any finite-sample divergence reflects differences in the bias-variance tradeoff and the inference channel rather than differential robustness to nonlinearity per se. This makes Extension 2 complementary to Extension 1: whereas Extension 1 isolates the bias channel under invertible dynamic misspecification, Extension 2 isolates the inference channel, since the nonlinear DGP generates non-Gaussian and serially dependent Wold innovations that affect the finite-sample coverage of LP and VAR differently. Critically, unlike the VARMA misspecification in Extension 1, the quadratic term is not approximable by higher-order linear VARs, which implies that the complexity-adjusted VAR sweep has no approximation advantage over VAR(4) in this setting.

The structural estimand is the best linear approximation to the true nonlinear IRF, $\theta_h^{\text{lin}} = \Psi_h B e_1$, where Ψ_h and B are defined with respect to the linear component of the DGP. Since both LP and VAR target this same linear estimand in population (Plagborg-Møller & Wolf, 2021), the comparison remains well-defined under nonlinear misspecification. Persistence is varied across $\rho \in (0.5, 0.95)$ as in the baseline and Extension 1, and sample sizes follow the baseline design $T \in \{100, 200, 240, 500\}$. The nonlinearity strength $\gamma \in \{0.3, 0.6\}$ is varied jointly with persistence and sample size. Both estimators are applied without modification to the nonlinearity-generated data, so that the misspecification is fully absorbed into the finite-sample performance of each estimator. Performance measures are identical to the baseline.

5.2 Simulation Results

6. Discussion (1.5-2 pages)

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